BEING GORENSTEIN IS PRESERVED AND REFLECTED BY DIVIDING BY A NONZERODIVISOR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The preservation of the Gorenstein property under dividing by a nonzerodivisor. Invoked at the start of the counted proof (source lines 4117-4118). Gives the forward implication.
- The vanishing criterion for the Gorenstein property through the Ext modules of the residue field. Invoked inside the counted proof (source lines 4119-4121 and 4131-4132). Translates between Gorensteinness and the eventual vanishing of those Ext modules in both directions.
- The adjunction between restriction and the internal hom along the quotient map. Invoked inside the counted proof (source lines 4122-4123). Computes the Ext modules of the residue field over the source in terms of those over the quotient.
- The computation of the internal hom out of the quotient as the quotient shifted by one. Invoked inside the counted proof (source lines 4129-4130). Turns the computed internal hom into the shift of the quotient, matching the criterion.

Source reference. Original lemma lemma-compute-for-effective-Cartier-algebraic: sources/stacks-dualizing/pinned-original-source.tex, source lines 2621–2621. Navigation pointer only: the counted text refers to the Stacks item lemma-compute-for-effective-Cartier-algebraic, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-divide-by-nonzerodivisor: sources/stacks-dualizing/pinned-original-source.tex, source lines 3383–3383. Navigation pointer only: the counted text refers to the Stacks item lemma-divide-by-nonzerodivisor,

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Source reference. Original lemma lemma-gorenstein-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 4063–4063. Navigation pointer only: the counted text refers to the Stacks item lemma-gorenstein-ext, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-right-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 2428–2428. Navigation pointer only: the counted text refers to the Stacks item lemma-right-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Let $f \in \mathfrak{m}$ be a nonzerodivisor. Set $B = A/(f)$. Then A is Gorenstein if and only if B is Gorenstein.*

Proof. If A is Gorenstein, then B is Gorenstein by Lemma lemma-divide-by-nonzerodivisor. Conversely, suppose that B is Gorenstein. Then $\mathrm{Ext}^i_B(\kappa, B)$ is zero for $i \gg 0$ (Lemma lemma-gorenstein-ext). Recall that $R\mathrm{Hom}(B, -) : D(A) \to D(B)$ is a right adjoint to restriction (Lemma lemma-right-adjoint). Hence

$$R\mathrm{Hom}_A(\kappa, A) = R\mathrm{Hom}_B(\kappa, R\mathrm{Hom}(B, A)) = R\mathrm{Hom}_B(\kappa, B[1])$$

The final equality by direct computation or by Lemma lemma-compute-for-effective-Cartier-algebraic. Thus we see that $\mathrm{Ext}^i_A(\kappa, A)$ is zero for $i \gg 0$ and A is Gorenstein (Lemma lemma-gorenstein-ext). □